

per cent stake in the company, an acknowledgement of the potential of these start-ups. “For an order to be delivered in 10 minutes, food must be ready and out of the kitchen in three minutes. People start ordering when they’re hungry. If they get tasty food [delivered] faster, they might actually pay a premium for the convenience they’re getting,” explains Mukunda Foods Co-founder & CEO Eshwar K. Vikas. “The No. 1 filter that’s used on Zomato and Swiggy is speed. That’s where we come in. We don’t regenerate the food; we cook it faster and people know the difference.”

According to Vikas, building a physical product (hardware) has many challenges because unlike a software product, changes cannot be made on the go. The product’s development is tedious, and usually takes four to five years with huge expenses. The other challenge for these tech firms is to get the taste of the food right because it’s more of a culinary art than a mechanical one. Hence, apart from the challenges of designing efficient hardware, taste and recipes play a major role. “The availability of quality talent in food tech, globally, is less, and you have to build the talent, which is time consuming,” he adds.

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THE INDIAN FOOD DELIVERY MARKET IS EXPECTED TO DOUBLE IN VALUE TO \$13 BILLION BY 2025, ACCORDING TO RESEARCH FIRM REDSEER

So, how do these robot chefs prepare your favourite *shahi paneer* once you order? A *shahi paneer* recipe can be made within three minutes using a Mukunda Foods kitchen robot called ‘Wokie’. First a chef makes the base gravy, the automated wok then adds paneer, cream and other spices in the perfect proportion at the touch of a button. And, your favourite *shahi paneer* is ready. Not just that, a whole list of recipes can be prepared. For example, another machine called ‘Misty’ can make *momos* from scratch, while ‘Dosamatic’ can churn out delectable *dosas* in two minutes. There are many other such machines these start-ups have designed, and many more are under development.

While cooking can be automated, what about taste? “We spend at least two to three years in getting the taste right. Nobody will buy the machines for its tech, if the food doesn’t taste nice. That’s why we work with a lot of in-house chefs,” says Vikas of Mukunda Foods. His machines, Vikas says, have solved the challenge of standardisation of taste, temperature and quality for QSRs and cloud kitchens. Other beneficiaries of their smart kitchen include brands such as Auntie Fung’s, edabba, Biggies Burger and Meraki, among others.

RoboChef, another company in the automated kitchens space, has taken the tech to the next level. Its kitchens are fully automated with minimal need of a chef to be present. Currently, its equipment is in use in about 50 kitchens in Chennai, Hyderabad and Singapore. It runs a multi-vessel system with digital recipes that are used to make biryanis and curries, and other dishes. If a QSR or cloud kitchen can shell out ₹1 crore for its solution, then churning out 20,000 portions of biryani in a few minutes is child’s play.

Not just KaaS start-ups, this trend has also led to a rise in cloud kitchens that are exclusively focussed on order delivery, and they are increasingly using technology and automation to reduce their daily production and delivery times. “A very natural behaviour is that you start thinking about food when you’re hungry. So, there are a lot of relevant use cases for 10-minute food delivery for consumers. Coming to the execution side of things, it’s a tricky problem to solve,” says Rohan Agarwal, Partner at RedSeer. “Deliveries are where it takes the most time, which means that the food preparation time needs to come down. That’s where these [kitchen tech] solutions will come in handy. They will be able to speed up the preparation, [and make the process] fail-safe, tech driven and consistent. It also helps you improve margins on the kitchen side because the cost of manual labour comes down,” he adds. According to RedSeer, cloud kitchens